Report

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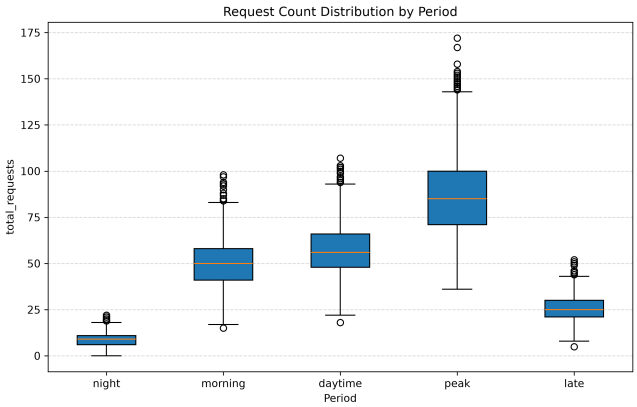


Figure 1: Box plots of text requests

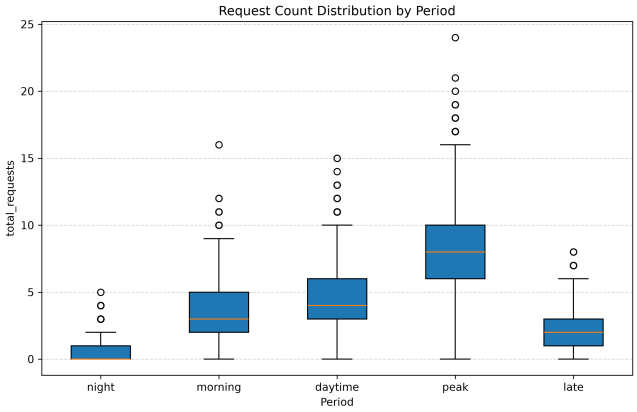


Figure 2: Box plots of image requests

0.1 Step 1

0.1.1 Drawing the Box Plots

The box plots for the two types of requests are presented in fig. 1 and fig. 2, respectively. It is evident that an analysis restricted to peak hours on weekdays alone cannot adequately illustrate the actual trends in request volumes.

0.1.2 Establish Poisson distribution

Poisson distributions are assumed for N_t and N_i, specifically,

$$P(N_t = k) = \frac{\lambda_t^k e^{-\lambda_t}}{k!},$$

$$P(N_i = k) = \frac{\lambda_i^k e^{-\lambda_i}}{k!}.$$

The parameters λ_t and λ_i can be estimated using the sample mean:

$$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} N_i.$$

The 95% confidence interval for the Poisson parameter λ can be approximated using the normal approximation:

$$CI_{95\%} = \left[\hat{\lambda} - 1.96\sqrt{\frac{\hat{\lambda}}{n}}, \hat{\lambda} + 1.96\sqrt{\frac{\hat{\lambda}}{n}} \right].$$

0.2 Step 2

0.2.1 Report for service time

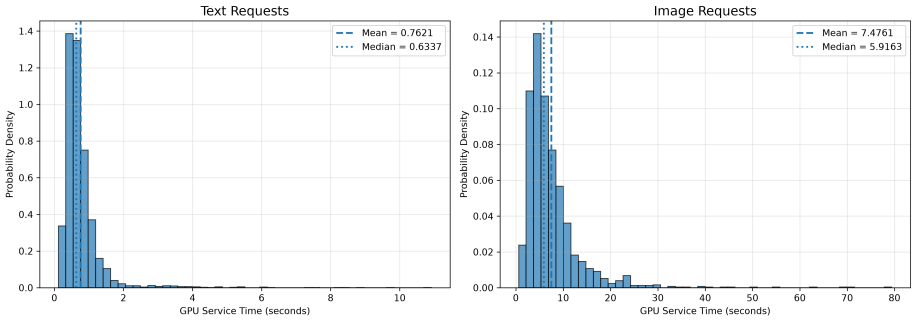


Figure 3: Histogram of two types of requests

Table 1: Descriptive Statistics of Text and Image Request Durations (seconds)

Type	Size	Mean (s)	Std (s)	Median (s)	95% (s)	99% (s)
Text	4000	0.762147	0.628502	0.633650	1.464310	3.528130
Image	1600	7.476124	6.247750	5.916300	17.529665	29.749451

0.2.2 Report for cold start service time

0.2.3 Establish Probability Model

0.2.4 Estimate Model Parameters